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Original Paper

Characterization of poly(ethylene glycol)-b-poly(ϵ -caprolactone) by two-dimensional liquid chromatography

Block copolymers of ethylene oxide and ϵ -caprolactone were synthesized by microwave-assisted polymerization of ϵ -caprolactone with polyethylene glycol monomethyl ethers as initiator. The samples thus obtained were characterized by two-dimensional liquid chromatography with liquid chromatography at critical conditions as the first and liquid exclusion adsorption chromatography as the second dimension. A full baseline separation of all oligomers could be achieved in both dimensions.

Keywords: Block copolymer / Caprolactone / Critical conditions / Microwave-assisted polymerization / Two-dimensional chromatography

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1 Introduction

Poly(ϵ -caprolactone) (PCL) is a promising material for biomedical applications owing to its good biocompatibility, permeability, and degradability. Block copolymers of CL with ethylene oxide (EO) have amphiphilic properties and are thus employed for formation of biocompatible micelles, as nanocarriers for drug delivery, for encapsulation of anticancer drugs, as thermosensitive biomaterials, and as porous scaffolds for tissue engineering techniques [1]. They are typically synthesized by ring-opening polymerization of ϵ -caprolactone with polyethylene glycol or its monoalkyl ethers in the presence of tin octoate as catalyst [2]. The polymerization has been performed using different catalysts, such as HCl in diethyl ether [3], lanthanum [4], titanium, calcium [5], and other metals, sometimes also with metal complexes [6]. The block copolymers thus obtained have a distribution in the length of both blocks – the ethylene oxide (EO) and the

caprolactone (CL) chain – and may contain impurities (mainly unreacted monomer and initiator as well as linear or cyclic homopolymers of caprolactone), which may seriously affect product properties. As many applications require well-defined products of a high purity, there is a pressing need for reliable characterization methods, which should allow an independent determination of the length of the individual blocks and the content of homopolymers and their molar mass distribution (MMD). Of course this cannot be achieved by simple methods. Basically, the independent determination of two distributions requires a two-dimensional separation. In practice, this can be done using 2-D HPLC or a combination of liquid chromatography with mass spectrometry, especially MALDI-TOF-MS [7–9].

In this study, chromatographic methods should be developed which allow a full separation of all oligomers in diblock copolymers within a reasonable time.

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Abbreviations: CAP, critical adsorption point; EO, ethylene oxide; FTD, functionality type distribution; LAC, liquid adsorption chromatography; LCCC, liquid chromatography at critical conditions; LEAC, liquid exclusion adsorption chromatography; LCLCA, liquid chromatography under limiting conditions of adsorption; LCLCS, liquid chromatography under limiting conditions of solubility; MMD, molar mass distribution; PEG-MMEs, polyethylene glycol monomethyl ethers; PCL, poly(ϵ -caprolactone); RI, refractive index; SEC, size exclusion chromatography

2 Mechanisms in polymer chromatography

Characterization of polymers can be accomplished by use of different modes of chromatography which separate according to different criteria. These mechanisms can be characterized by the interaction parameter c , which has been introduced by De Gennes [10–15]

2.1 Size exclusion chromatography

Size exclusion chromatography (SEC) separates according to the hydrodynamic volume of the polymer in solution, *i.e.* the molecular size of the entire molecule. In

ideal SEC the interaction parameter is strongly negative, and interactions with the stationary phase should be absent. This can be achieved by using a thermodynamically good solvent (a strong displacer) for the polymer, which effectively suppresses enthalpic interactions with the stationary phase [16]. By appropriate calibration one can determine the molar mass distribution (MMD). As SEC is typically performed in single eluents, dual detection can be used to determine the chemical composition of (binary) copolymers along the MMD [17–20]. SEC is, however, not capable of discriminating copolymers and mixtures of homopolymers.

2.2 Liquid adsorption chromatography

Liquid adsorption chromatography (LAC) separates according to the number of groups, which may interact with the stationary phase [21]. It is governed by enthalpic effects. In LAC the interaction parameter is positive. Therefore, elution volumes V_e are larger than the void volume V_0 . In principle, LAC can be run with isocratic or gradient elution. Gradient elution is often required, because retention increases exponentially with the degree of polymerization. Solvent gradients and temperature gradients can be used for a separation according to chemical composition [22–24]. In the range of lower molar mass (up to several thousands) gradient elution allows a baseline separation of the individual oligomers [25–27].

2.3 Liquid chromatography under critical conditions

Liquid chromatography under critical conditions (LCCC) is a technique, which allows a separation of polymers according to structural units different from the repeat unit. In each system, there may be a mobile phase composition and temperature at which the polymer chain becomes “chromatographically invisible”. At this so-called critical adsorption point (CAP) for a given repeat unit, entropic and enthalpic effects compensate each other, and the interaction parameter of this repeat unit equals zero. As a result polymeric homologues are eluted at the same elution volume irrespective of their molar mass. At the CAP for unit A, the corresponding block A becomes chromatographically invisible [28–31]. This allows a separation according to structural units different from the repeat unit, e.g. the end groups or a block consisting of the repeat unit B [29, 32–34]. At the CAP for A, the other block B may elute in SEC or LAC mode. Consequently, a block copolymer A-B elutes earlier (in SEC mode) or later (in LAC mode) than the (non-functional) homopolymer A, which elutes at the void volume. The situation with an excluded block B is useful for higher polymers; in the case of an adsorbing group B one may deter-

mine the functionality type distribution (FTD). It is also possible to separate linear and cyclic species [35].

2.4 Liquid exclusion adsorption chromatography

In A-B block copolymers or functional polymers, the interaction parameters of the corresponding units may have the opposite sign ($c_A < 0$ and $c_B > 0$). In this case, a separation of the individual oligomers of the excluded block A may be achieved, which elute in SEC order, but at elution volumes much larger than the void volume (depending on the interaction parameter of the adsorbing block B). This technique is called liquid exclusion adsorption chromatography (LEAC). It allows a baseline separation of individual oligomers under isocratic conditions. As LEAC is performed under isocratic conditions, a refractive index (RI) detector may be used, which allows accurate quantification in the case of samples without chromophoric groups. In practice, LEAC can only be applied to copolymers in which the adsorbing block is uniform, as is the case in nonionic surfactants. For poly-disperse adsorbing blocks, the individual series will overlap. In the second dimension of a two-dimensional separation, LEAC is very useful if such fractions can be obtained from LCCC as first dimension [36–39].

2.5 Liquid chromatography under limiting conditions of solubility

Another interesting approach, which allows a separation of homopolymers and copolymers, is called liquid chromatography under limiting conditions of solubility (LCLCS) [40] or liquid chromatography under limiting conditions of adsorption (LCLCA) [41]. A polymer mixture dissolved in a good solvent for both homopolymers is injected in a mobile phase of a different composition, which is also a good solvent for polymer A but a poor solvent (or even a non-solvent) for polymer B. In this case, the homopolymer A will escape from the zone of the strong eluent, while the homopolymer B and the copolymer AB will travel with the solvent peak [42–44].

3 Approaches in two-dimensional separations

The mechanisms mentioned above may be combined for two-dimensional separations. Chromatography may also be coupled with soft ionization techniques like electrospray ionization (ESI) and matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (MALDI TOF-MS) as second dimension [45–50].

In the case of two-dimensional liquid chromatography (2-D LC) the dimensions can be coupled in different ways.

The simple and safest way is a semi-preparative separation as the first dimension, and subsequent analysis of the fractions thus obtained by a different mechanism in the second dimension [51, 52]. This is even possible if the mobile phases used in both dimensions are not compatible. The main drawback is that very careful handling of the fractions is required in order to avoid contamination.

This can be avoided by the full adsorption–desorption technique (FAD) [38, 39, 53–55]: in this case fractions are trapped on a short column, on which they are reconcentrated. The narrow zones thus obtained are subsequently injected into the second dimension by switching to a stronger eluent.

A rather new approach is called comprehensive chromatography [56–58], in which fractions are automatically transferred from the first to the second dimension. Such separations have gained increasing attention in the last years as they can be performed automatically and yield contour maps which may serve as a fingerprint of a polymer sample. This approach has, however, one major drawback, which is typically not mentioned in the literature: it can only be applied under special preconditions, as will be discussed later on, and it works under far from optimal conditions in both dimensions.

In principle, the following combinations may be used in two-dimensional separations: LCCC $\times$ SEC, SEC $\times$ LCCC, LCCC $\times$ LCCC, LAC $\times$ LCCC, and LAC $\times$ SEC.

Of course, there are some limitations: if the mobile phases in the first and second dimension are not compatible, a direct transfer of fractions from the first to the second dimension is impossible.

In 2-D LC one has to keep in mind the following considerations. If an aqueous mobile phase is used in the first dimension (e.g. LAC or LCCC on a RP column), it is dangerous to inject the fractions on a typical SEC column running in an organic (non-polar) mobile phase such as toluene, chloroform, etc.: this may destroy the column.

On a normal-phase column (e.g. plain silica) there will be no retention at all for fractions from an RP separation (in an aqueous mobile phase), as water deactivates the polar surface of the stationary phase.

A fraction dissolved in a strong eluent injected on a LAC column running in a weaker eluent will often travel with the solvent peak (as is the case in LCLCA).

In the classical approach (with collection of fractions) all combinations can be applied, as one may remove the mobile phase of the first dimension from the fractions and dissolve the dried fractions in the mobile phase used in the second dimension. This is, however, laborious and may lead to a contamination of the fractions.

Even if direct transfer between the dimensions is possible, there are still some other problems in comprehensive chromatography which require compromises:

A chromatogram in the second dimension must be finished in the time required to fill the sample loop in the first dimension. This means, that in the second dimension small columns must be used at a very high flow rate (which is limited by the back pressure). On the other hand, the first dimension must be run at a very low flow rate. Evidently, both are far from the optimum in the van Deemter equation, especially in the first dimension, as the plate height increases dramatically at very low linear velocity.

Another question concerns the size of the sample loops. Injecting too large a volume on a small column will dramatically decrease the efficiency of the second dimension. On the other hand, using smaller loops will result in a large number of chromatograms in the second dimension, most of which do not contain anything.

Moreover, the small amounts injected in the second dimension may cause serious detection problems. After all, such a separation takes a lot of time and solvents. This may be illustrated by the following example.

Let us assume a separation in the first dimension, in which the last peak appears at an elution volume of 30 mL. At a flow rate of 0.1 mL/min, this will require 5 h. Using sample loops of 200 μ L volume, this will mean 150 chromatograms in the second dimension. As the time required to fill the loop in the first dimension is 2 min, a separation in the second dimension must be finished in this time, which may require flow rates of 4–5 mL/min on rather short columns. A sample volume of 200 μ L is, however, very large for the small columns used in the second dimension. Consequently, one may not expect good resolution in such a separation.

There is, however, a simple and convenient alternative, especially for samples where discrete peaks can be obtained in the first dimension. This is the case for functional polymers and block copolymers with rather short blocks (typically up to 20–30 repeat units). If the mobile phases of the two dimensions are compatible, one may run the first dimension on a semipreparative column at optimized conditions with a fraction collector, fill the fractions thus obtained in autosampler vials, and run the second dimension at a reasonable flow rate under optimized conditions. In this case, there is no need to run very fast separations: in practice, a copolymer containing 10 oligomer fractions (regarding the number of repeat units A) can be well separated in less than 30 min. Filling the fractions into autosampler vials may take just a few minutes, and each of the separations in the second dimension perhaps also 20 min or less. Consequently the total analysis time may be less than 4 h. This approach has the advantage that the fractions can also be analyzed by MALDI-TOF-MS or other spectroscopic techniques. Moreover, one may repeat the separation in the second dimension under different conditions.



In the present paper, we have applied this technique to the separation of EO-CL block copolymers. The goal of this study was to achieve a full separation of all oligomers in a reasonable time.

4 Experimental

4.1 Materials

Polyethylene glycol monomethyl ethers (PEG-MMEs) with average molar masses of 350, 550, 1100, monoethylene glycol methyl ether, diethylene glycol methyl ether, triethylene glycol methyl ether, and ϵ -caprolactone were purchased from Fluka (Buchs, Switzerland), tin(II) octoate from Sigma–Aldrich. The solvents used in chromatography (water, acetone, acetonitrile, and methanol) were all of HPLC grade and purchased from Carl Roth GmbH, Karlsruhe, Germany.

4.2 Polymerization

Caprolactone was polymerized using tin(II) octoate catalyst. Different PEG-MMEs, methanol, and monoalkylated glycols were used as initiators. All polymerizations were performed in bulk under an argon atmosphere. The catalyst was added to the initiator, then the monomer was added. Polymerizations were carried out under microwave heating [59] at 140°C for 30 min in an Emrys ‘Optimizer’ Microwave instrument (Biotage, Uppsala, Sweden). Under these conditions complete conversion of the monomer was achieved, as was confirmed by HPLC.

4.3 Chromatography

These investigations were carried out on two different chromatographic systems.

Preparative chromatography was performed at a temperature of 25°C. The mobile phase was delivered by a JASCO 880 PU pump (Japan Spectroscopic Company, Tokyo, Japan) at a flow rate of 2.0 mL/min. Samples were injected manually using a Rheodyne 7125 injection valve (Rheodyne, Cotati, CA, USA) equipped with a 500 μ L loop. A semi-preparative column was used for all measurements: Spherisorb ODS2 (25 $\times$ 300 mm, 5 μ m, 80 Å) from PhaseSep (Deeside, UK). The separations by LCCC in the first dimension were performed using a DDS 70 density detection system (CHROMTECH, Graz, Austria) which has been developed by our group. Data acquisition and processing was carried out by using a CHROMA software package. A LCD 201 RI detector (Sicon Analytics, Germany) was combined with the DDS 70. Fractions were collected using an Advantec SF 2120 fraction collector (Advantec, Dublin, CA, USA). The column and density cell was placed in a thermostatted box, in which a temperature of 25.0°C was maintained for all measurements using a Lauda RM6

thermostat (Lauda, Lauda-Königshofen, Germany). Mobile phases for LCCC were mixed by mass and vacuum degassed, their composition was controlled by density measurement using a DMA 60 density meter equipped with a DMA 602 M measuring cell (A. Paar, Graz, Austria).

The separations in the second dimension were performed on an Ultimate 3000 gradient system, consisting of a DGP-3600A pump, a SRD-3600 solvent degasser, a TCC-3000 column thermostat, a WPS-3000SL autosampler, all from Dionex (Germerink, Germany), a PL-ELS 2100 evaporative light scattering detector (Polymer Laboratories/Varian, Church Stretton, Shropshire, UK). Data acquisition and processing was performed using Chromeleon software (Dionex, Germerink, Germany). All separations were performed on a Synergi Fusion RP column (silica-based dodecyl phase; 250 $\times$ 4.6 mm; particle diameter = 4 μ m; nominal pore size = 80 Å, from Phenomenex, Torrance, CA, USA) at a flow rate of 0.5 mL/min. The injected sample volumes depended on the total volume of the fraction from the first dimension in order to inject comparable sample sizes. The mobile phases in the second dimension were mixed by the gradient pump, so their compositions are given in volume percent. The chromatograms shown in this paper were run with isocratic elution, unless mentioned otherwise.

The following gradient was used: start 40% acetone–water, in 15 min to 70% acetone, 30 min to 90% acetone, 9 min constant 90%, then back to 40% acetone in 1 min.

5 Results and discussion

In order to have reliable reference materials for the development of the analytical methods, we have synthesized a number of model polymers by microwave-assisted polymerization of ϵ -caprolactone (CL) using different initiators such as water, methanol, mono-, di- and tri-ethylene glycol monomethyl ether and polyethylene glycol monomethyl ethers (MW 350–1100).

These products were characterized as described in a previous communication. The overall molar mass was determined by SEC, and the functionality type distribution (FTD) by LCCC: on reversed phase columns, a CAP for CL was found in THF–water (80:20). Under these conditions the copolymer elutes in SEC mode and can thus be separated from the PCL homopolymer. This was also performed on a semipreparative scale, and the fractions thus obtained were characterized by SEC and MALDI-TOF-MS.

The aim of the present study was, however, a full separation of all oligomers. Therefore we chose a different approach. At the CAP for the more polar EO unit on a RP column, the nonpolar PCL should elute in LAC mode. If a separation of the individual oligomers could be achieved, these fractions, which contain polymers uni-

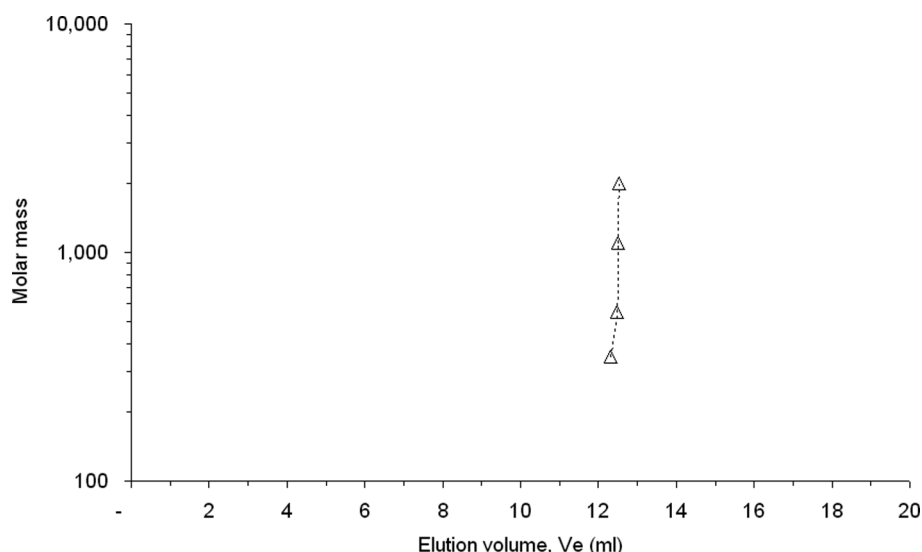


Figure 1. PEG-MMEs on Spherisorb ODS (prep) in 69.12 wt-% acetonitrile–water.

form in the number of CL units, could be separated in the second dimensions according to the number of EO units.

A separation of the non-polar PCL under critical conditions for the polar EO unit can only be possible on reversed phases. On typical RP columns, a CAP for EO exists in 85–95 wt-% methanol–water [60]; in acetone–water two critical points have been observed: at 26–30 wt-% and at 90–93 wt-% (the latter exists only on silica–based stationary phases) [60, 61]. In 90 wt-% methanol–water low molecular PCL samples could be reasonably separated, but higher PCLs are not soluble in such a mobile phase. PCL is also insoluble in 30% acetone, and in 90% acetone the retention of PCL is not sufficient. On RP phases, critical conditions for EO exist also in acetonitrile–water. We have found that PEG and PEG-MME elute close to the CAP in quite a wide range of mobile phase composition: in 69 wt-% acetonitrile–water PEG is sufficiently close to critical conditions (Fig. 1).

In such a mobile phase, PCL is sufficiently soluble, and the lower PCL oligomers can be separated quite well by IAC (Fig. 2).

Evidently, the retention of the individual oligomers decreases with the number of EO units attached to the PCL chain, which is surprising as in such a mobile phase PEG-MME elutes even slightly on the adsorption side. The reason for this behaviour could be that the pores of this column (80 Å) may be somewhat too small. Another possibility explanation might be that a larger critical block shields the adsorbing block, which is not the case if the individual blocks are not compatible with each other. In the near future a detailed study will be performed to elucidate the reason for this phenomenon.

As can be seen in Fig. 2, a reasonable separation according to the number of CL units can be achieved if the EO block has a molar mass below 1000.

On this column, we have fractionated a block copolymer synthesized with PEG-MME 350 as initiator and an average degree of polymerization $n = 5$. This sample was chosen for all two-dimensional separations.

In the following text, diblock copolymers initiated with polydisperse PEG-MME will be referred to as MeO-(EO) m -(CL) n . Fractions from LCCC will be named according to their number of CL units: MeO-(EO) m -(CL)3, etc. The polymer initiated with tri-EG monomethyl ether will be called MeO-(EO)3-(CL) n , and so on.

As can be seen in Fig. 3, a baseline separation of 9 peaks could be achieved, corresponding to polymers containing 1 to 9 CL units. In this figure, fraction limits are indicated by vertical lines. The empty fractions were discarded. The fractions containing the series with 3 to 9 CL units should then be separated in the second dimension according to the number of EO units.

As mentioned above, this can be done in different ways. The most convenient situation would exist if the fractions could be injected as such in the second dimension. Obviously, the injection of a solution containing 24% water on a normal phase column does not make sense, as this would deactivate the silica surface.

A better approach would be LEAC: in a mobile phase, where CL is strongly adsorbed and EO elutes in SEC mode, one may expect a separation according to the number of EO units, wherein the smallest oligomer appears last, as has already been shown in the analysis of non-ionic surfactants [37]. As in SEC, elution volumes in LEAC decrease with increasing number of repeat units, but the dependence is different from that in SEC.

We have thus analyzed fractions from LCCC on a Synergi Fusion RP column in acetone–water mobile phase in a range of composition in which the interaction parameter of the EO unit is negative, i.e. between 40 and 90% acetone. In the second dimension we have adjusted

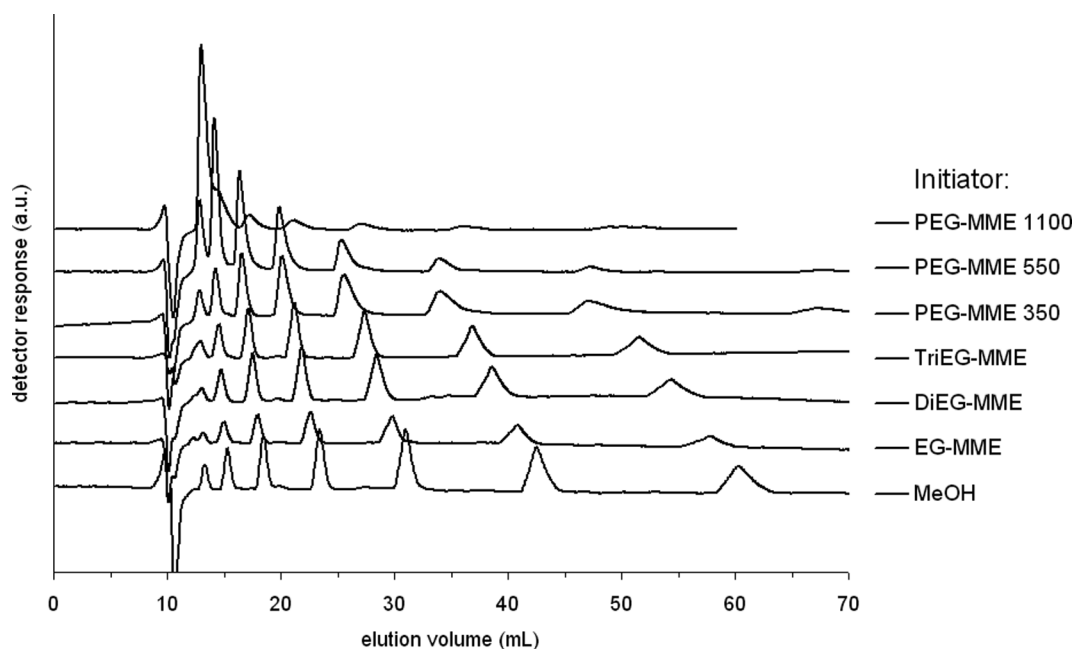


Figure 2. LCCC separation of PCL synthesized with different initiators (average degree of polymerization: $n = 5$) on Spherisorb ODS ($5\ \mu\text{m}$, $80\ \text{\AA}$, $250 \times 10\ \text{mm}$) in 69.12 wt-% acetonitrile–water. Detection: density.

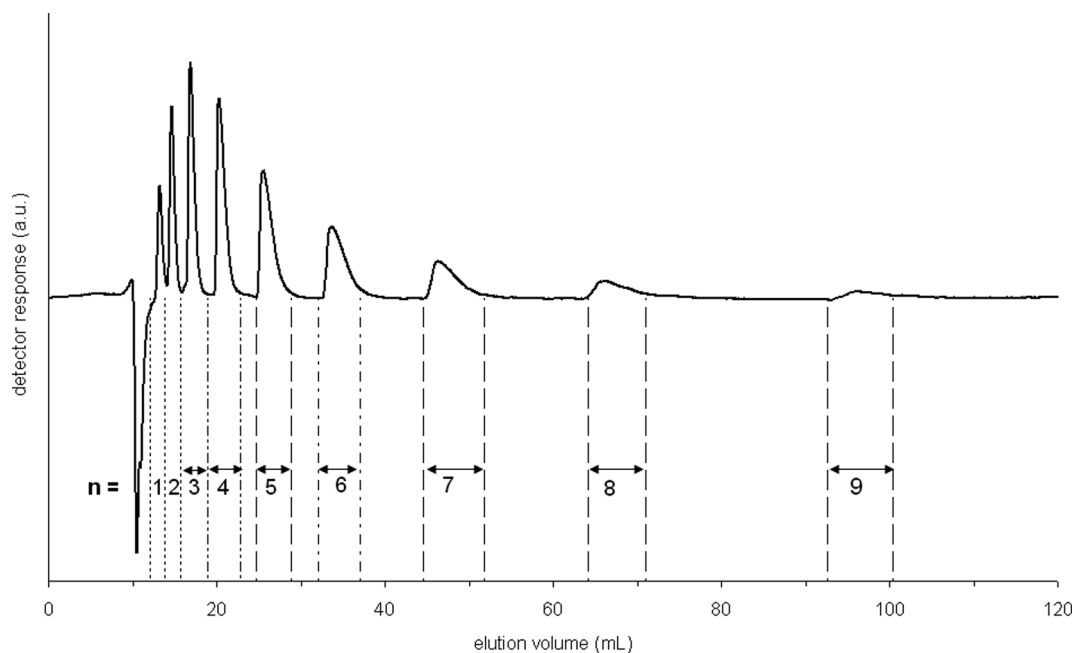


Figure 3. Fractionation of the sample initiated with PEG-MME 350 (see Fig. 1) by LCCC on Spherisorb ODS (prep) in 69.06 wt-% acetonitrile–water. Detection: density.

the mobile phase composition using a gradient pump, hence the composition of the mobile phase is always given in vol-%.

As can be seen in Fig. 4, all oligomers of the fraction with 7 CL units (containing 3–13 EO units) could be baseline resolved in a mobile phase containing 68% acetone.

The individual peaks were identified by comparison with samples based on a uniform initiator (in this case TriEG-MME). The oligomer peak used for identification is labelled by a rectangle.

The series with less CL will, of course, require a different mobile phase composition for a sufficient resolution.

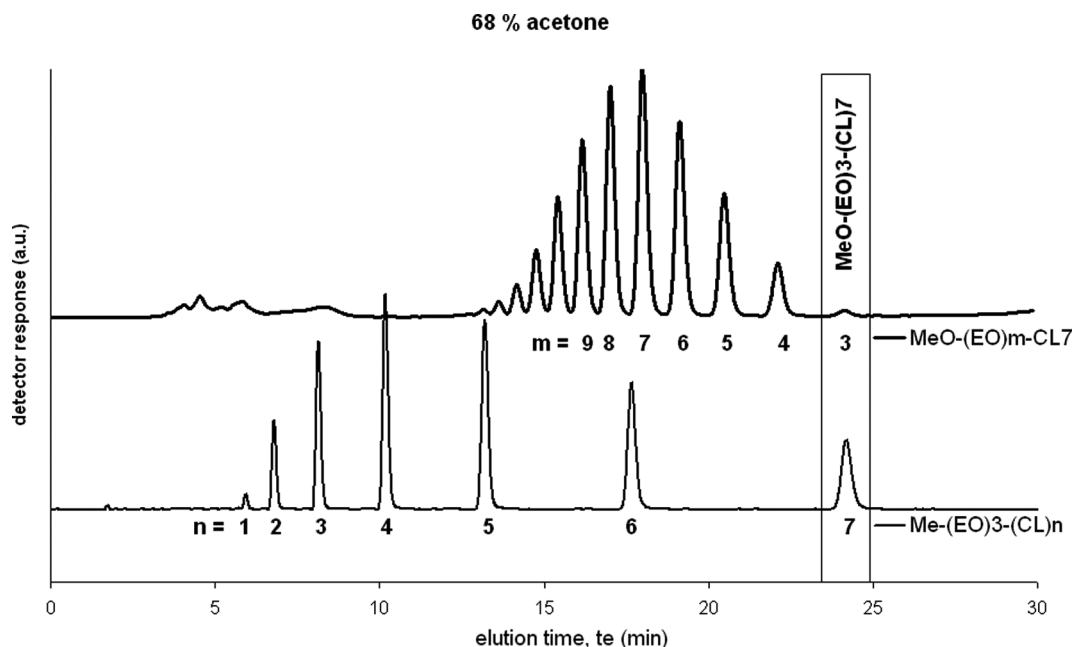


Figure 4. LEAC of a fraction from LCCC (Fig. 3) containing 7 CL units and a polymer initiated with Tri-EG-MME, as obtained on Synergi Fusion RP (250 × 4.6 mm, 5 μm) in 68 vol-% acetone–water. Detection: ELSD.

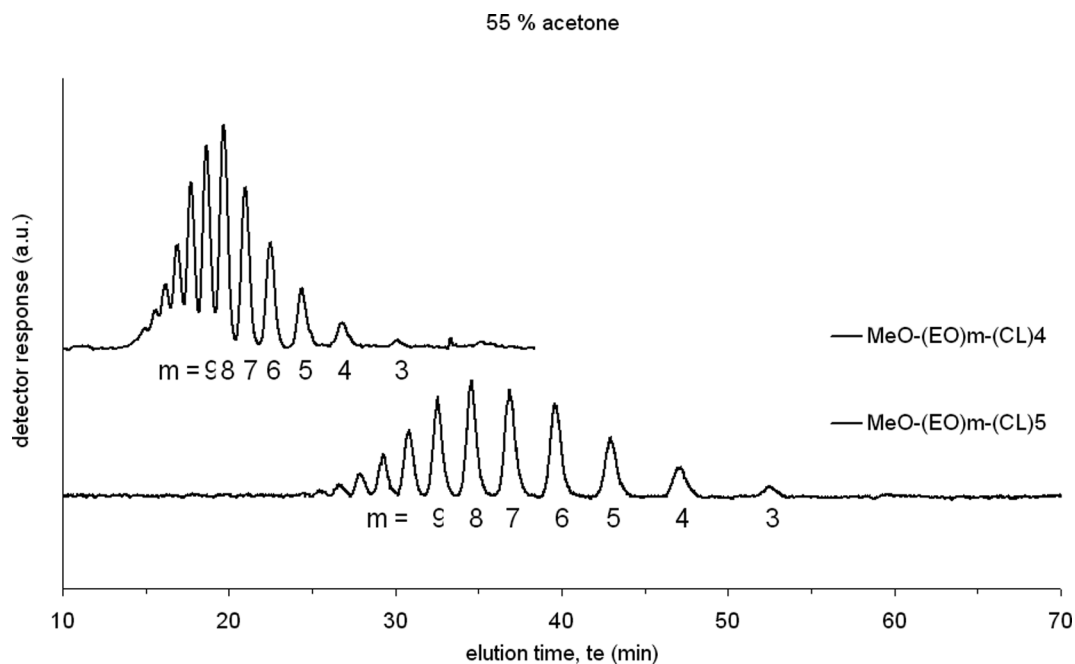


Figure 5. LEAC of two fractions from LCCC (Fig. 3) containing 4 and 5 CL units, respectively. Synergi Fusion RP, 55 vol-% acetone–water. Detection: ELSD.

Figure 5 shows the separation of the series containing 4 and 5 CL units, respectively, in 55% acetone. Both series slightly overlap in this chromatogram.

For the fractions containing a larger number of CL units, the individual series of peaks strongly overlap, as can be seen in Fig. 6. Consequently, LEAC can only be

applied to fractions which are uniform with respect to the number of CL units. Such fractions can easily be obtained using LCCC (at the CAP for EO) in the first dimension.

In principle, LEAC is typically performed with isocratic elution. It should, however, also be possible to run a sep-

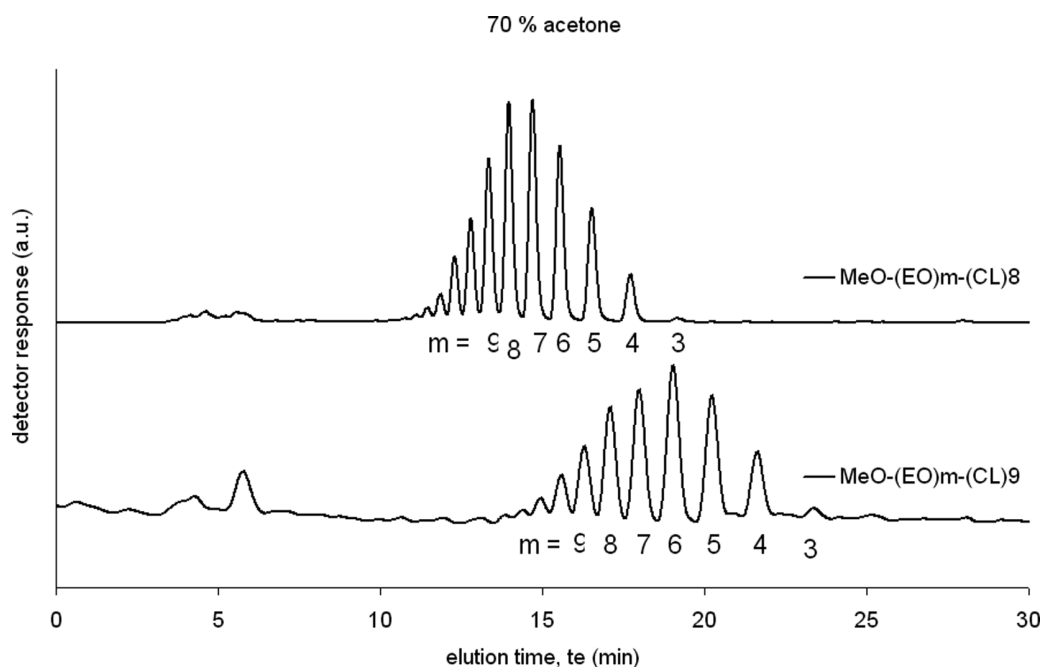


Figure 6. LEAC of two fractions from LCCC (Fig. 3) containing 7 and 8 CL units, respectively. Synergi Fusion RP, 70 vol-% acetone–water. Detection: ELSD.

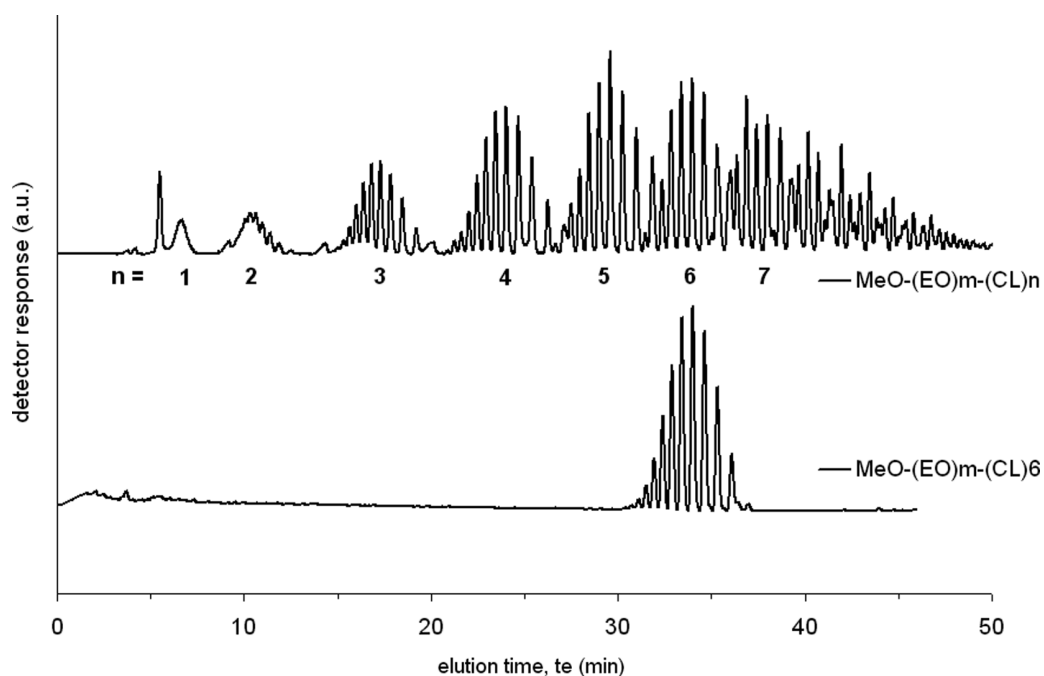


Figure 7. Gradient chromatogram of the original sample and a fraction from LCCC (Fig. 3) containing 6 CL units, as obtained on Synergi Fusion RP. Detection: ELSD.

aration with gradient elution. As can be seen in Fig. 7, the individual oligomers of a series with 6 CL units can be quite well resolved and identified under such conditions.

With the raw sample, however, several overlapping series of peaks are obtained with the same gradient, as

can be seen in Fig. 7. While the series containing a smaller number of CL units are not completely resolved, but separated from each other, the opposite is the case with the series containing a larger number of CL units. These would be baseline resolved, but the individual series strongly overlap.

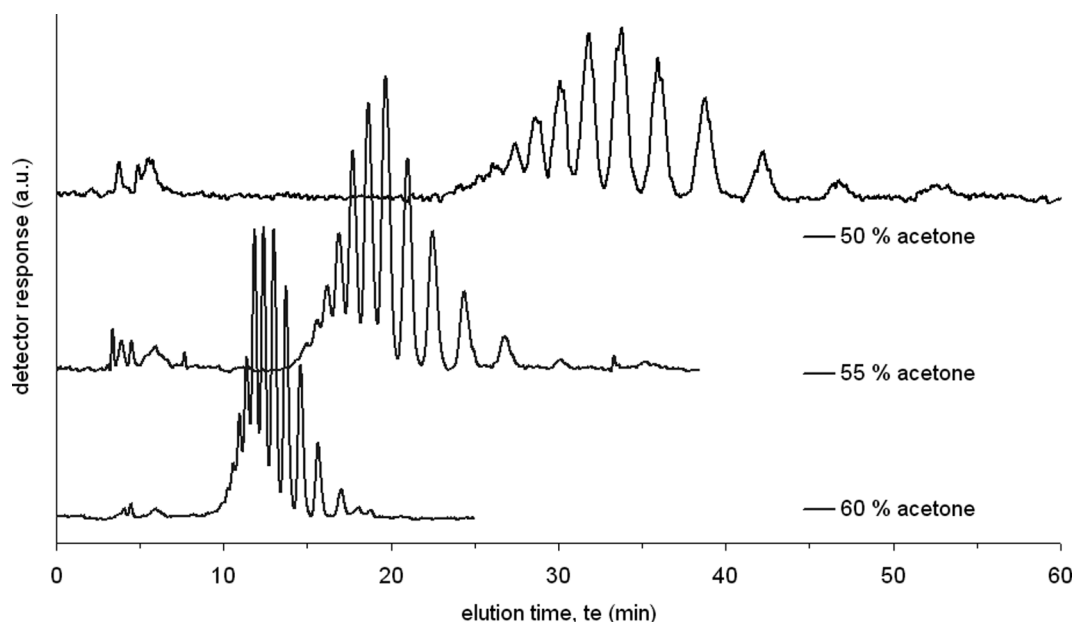


Figure 8. LEAC of a fraction from LCCC (Fig. 3) with 4 CL units, as obtained on Synergi Fusion RP in different mobile phase compositions. Detection: ELSD.

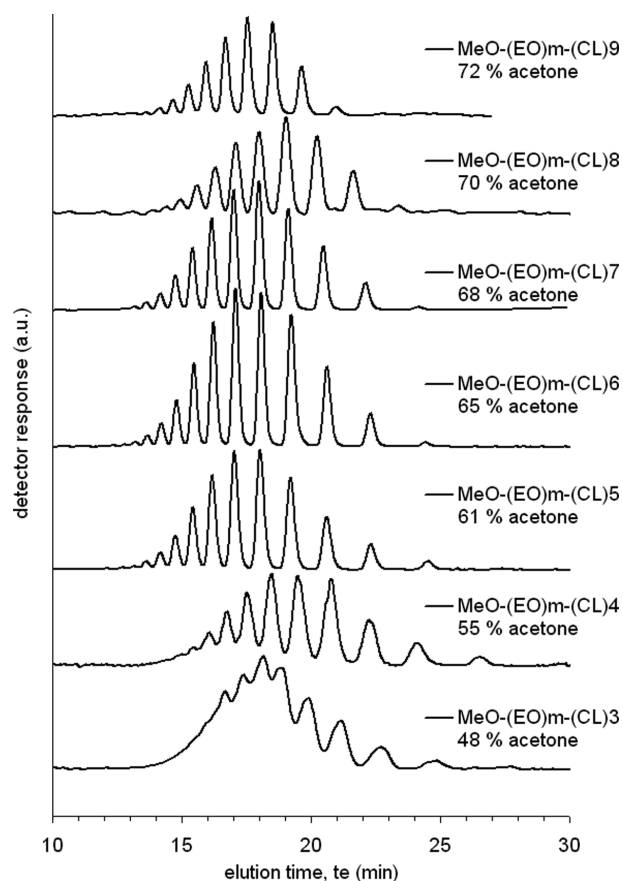


Figure 9. LEAC of different fractions from LCCC (Fig. 3), as obtained on Synergi Fusion RP in the optimal mobile phase composition.

In an appropriate mobile phase composition, the same resolution should be possible for all series (with a different number of CL units), as long as the interaction parameter of the EO unit is still sufficiently negative. As can be seen in Fig. 8, retention of the individual oligomers and thus resolution increases with decreasing acetone content of the mobile phase.

In an appropriated composition of the mobile phase, the oligomers of the individual fractions can indeed be baseline separated in a reasonable time and identified.

In the mobile phase of optimized composition, the individual oligomers of all fractions with 4 or more CL units can be resolved to the baseline, as can be seen in Fig. 9. All fractions show the same distribution of the EO block. Only for the fraction with 9 CL units, the lowest oligomers seem to be missing. The reason for this may be too narrow fraction limits in the first dimension.

For the fractions with 3 or less CL units, the resolution is not so good. This may be because a sufficient retention of the hydrophobic part is only achieved in a mobile phase composition quite close to the CAP for the EO unit, which is the limitation of the LAC separation.

6 Concluding remarks

Lower diblock copolymers of ethylene oxide and ϵ -caprolactone can be characterized very well by off-line two-dimensional liquid chromatography with LCCC as the first and LEAC as the second dimension. In an appropriate mobile phase composition, more than 10 oligomers can be baseline separated in the second dimension. On a

25 cm column, this takes typically about 30 min at a flow rate of 0.5 mL/min. For 6–8 fractions from the first dimension (where a chromatogram takes 60–120 min), the overall analysis time is typically less than 5–6 h. On a monolithic column, where higher flow rates can be applied, the time required for a chromatogram in the second dimension could be considerably reduced. As the fractions from the first dimension can be directly injected into the second dimension, it should also be possible to run a similar separation with on-line coupling of the dimensions. Anyway, the overall time required for the analysis will not be shorter than in the off-line mode. The scope and limitations of comprehensive chromatography for such a separation problem will be the subject of another communication.

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